

AnalystLab Africa

Machine Learning Internship Program – Batch B

Week 3 Summary Report

Machine Learning Fundamentals

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Dataset Used: Titanic Survival Dataset

Submission Deadline: Sunday, 21 June 2026, 11:59 PM WAT

1. What I Learned This Week

1.1 Supervised vs Unsupervised Learning

This week I applied both supervised and unsupervised machine learning to the Titanic dataset. In supervised learning, I used the known survival labels (0 = did not survive, 1 = survived) to train a K-Nearest Neighbours classifier. The model learned that features such as passenger class, sex, and age strongly predicted survival outcomes.

For unsupervised learning, I stripped the survival labels and applied K-Means clustering (k=2) to passenger age, fare, and class. The algorithm discovered two natural groupings — broadly corresponding to higher-fare, upper-class passengers and lower-class, lower-fare passengers — without ever seeing the survival column.

Property	Supervised	Unsupervised
Data	Labelled (survival 0/1)	Unlabelled (no survival column)
Goal	Predict survival outcome	Group similar passengers
Algorithm	KNN, Decision Tree	K-Means (k=2)
Titanic insight	Sex + class were strongest predictors	Clusters aligned with fare/class bands

1.2 Train / Test Split

I performed an 80/20 stratified split on the 891-passenger dataset, yielding 712 training samples and 179 test samples. Stratification ensured that the ~38% survival rate was preserved in both subsets, preventing a biased evaluation. This step is critical before any model is deployed — it simulates how the model will behave on future, unseen passenger data.

1.3 Overfitting and Underfitting

Using Decision Trees at depths 1 through 19 revealed the classic bias-variance tradeoff on Titanic data. At depth 1–2 the model underfits: it can only split on one or two features and misses the complex interaction between sex, class, and age. Around depth 5 the model generalises best. Beyond depth 8, the training accuracy climbs toward 100% while test accuracy plateaus or slightly falls — a clear sign of overfitting.

Titanic is noisier than Iris, so even the best-fit model achieves around 78–80% test accuracy, illustrating that real-world data always contains irreducible error that no model complexity can eliminate.

1.4 Model Evaluation Metrics

I evaluated the KNN classifier (k=5) using accuracy, precision, recall, F1-score, and a confusion matrix. On Titanic, recall is especially important: a false negative means predicting a passenger did not survive when they actually did — a serious error in any life-critical system.

- Accuracy — proportion of all correctly classified passengers
- Precision — of all passengers predicted to survive, how many truly survived
- Recall — of all passengers who actually survived, how many did the model identify

- F1-Score — harmonic mean of precision and recall; handles class imbalance better than accuracy alone
- Confusion Matrix — four-cell breakdown revealing true positives, true negatives, false positives, and false negatives

2. Challenges Encountered

2.1 Handling Missing Data Before Modelling

The Titanic dataset contains significant missing values, especially in the 'age' column (~20% missing). KNN cannot accept NaN values natively, so I had to apply median imputation before splitting. This taught me that data cleaning must happen before splitting to avoid data leakage — or that imputation should ideally be wrapped inside a pipeline applied only to training data.

2.2 Class Imbalance

The dataset is imbalanced: ~62% did not survive vs ~38% survived. A naive model that always predicts 'did not survive' would achieve 62% accuracy — making raw accuracy misleading. This reinforced the importance of examining precision and recall separately for each class, not just the overall accuracy figure.

2.3 Interpreting K-Means on Social Data

Unlike the Iris dataset, where clusters map neatly to biological species, Titanic clusters reflect socioeconomic groupings rather than survival directly. This was a valuable lesson: unsupervised clustering finds natural data structure, but interpreting what that structure means requires domain knowledge and careful cross-referencing with other variables.

3. Key Insights Gained

- Real-world datasets like Titanic are messier than textbook ones — missing values, class imbalance, and correlated features require extra preprocessing before any model can be trained.
- Accuracy alone is a poor metric when classes are imbalanced; precision and recall together give a more honest picture of model performance.
- The train/test split is non-negotiable — without it, there is no way to know whether a model has truly learned or simply memorised.
- Overfitting on Titanic happens faster than on Iris, confirming that noisier datasets require stronger regularisation or simpler models.
- K-Means on social data clusters by demographic similarity (fare/class), which aligns partially — but not perfectly — with survival, showing that unsupervised methods reveal hidden patterns but require human interpretation.
- Comparing two datasets (Iris vs Titanic) in the same week highlighted how dataset characteristics (noise, class balance, feature type) directly affect model performance and the choice of evaluation strategy.

4. Summary of Topics Covered

Topic	Titanic Application	Key Takeaway
Supervised Learning	Predict survival (0/1) with KNN	Sex & class are strongest survival predictors

Unsupervised Learning	K-Means clusters on age/fare/class	Clusters align with socioeconomic bands
Train/Test Split	712 train / 179 test, stratified	Preserves class ratio; honest evaluation
Overfitting	Decision Tree depth > 8	Train acc rises; test acc plateaus
Underfitting	Decision Tree depth 1–2	Too simple; misses sex-class interaction
Accuracy	KNN ~73–75%	Misleading with imbalanced classes
Confusion Matrix	TP/TN/FP/FN breakdown	Reveals cost of false negatives
Precision & Recall	Recall critical for survivor detection	High recall = fewer missed survivors

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